

- **Definition and function of inventory**
- Inventory is a **stock** of any items kept by an organization to meet internal or external **customer demand**.
- **Purchased part inventories permit activities of purchasing and supply personnel to be planned and conducted some what independently of shop production operations.**
- By the same token, these inventories allow additional **flexibility** for suppliers in planning, producing, and delivering an order for a given part

Types of inventories

- **Raw Materials.** These items include petroleum products, coal, metals and non-metals.
- It may include agricultural materials like soybeans and cotton
- **Semi-finished Products, Parts and Components**
- These include all those items, which enter in to final production.
- They are single component, sub assembly, assembly, sub system and system.
- **Finished Products**
- The organizations purchase the finished parts either for their internal use or resale.
- **Maintenance, Repair and Operating Items (MRO)**
- These items include any materials or products that does not enter production line but are essential for running the industry or business

PURPOSES OF INVENTORY

- **1. To maintain independence of operations.** A supply of materials at a work center allows that center flexibility in operations.
- For example, because there are costs for making each new production setup, this inventory allows management to reduce the number of setups.
- **2. To meet variation in product demand.** If the demand for the product is known precisely, it may be possible (though not necessarily economical) to produce the product to exactly meet the demand.
- **3. To allow flexibility in production scheduling.** A stock of inventory relieves the pressure on the production system to get the goods out.
- This causes longer lead times, which permit production planning for smoother flow and lower-cost operation through larger lot-size production.

- **4. To provide a safeguard for variation in raw material delivery time.**
- When material is ordered from a vendor, delays can occur for a variety of reasons:
 - **a normal variation in shipping time**
 - **a shortage of material at the vendor's plant causing backlogs**
 - **an unexpected strike at the vendor's plant or at one of the shipping companies**
 - **a lost order, or a shipment of incorrect or defective material.**
- **5. To take advantage of economic purchase order size.** There are costs to place an order:
 - labor,
 - phone calls
 - Typing
 - postage, and so on.
- Therefore, the larger each order is, the fewer the orders that need be written. Also, shipping costs favor larger orders the larger the shipment, the lower the per-unit cost.

- **6. Many other domain-specific reasons.** Depending on the situation, inventory may need to be carried. For example, in-transit inventory is material being moved from the suppliers to customers and depends on the order quantity and the transit lead time.

Dependent and independent demand

- In independent demand, the demands for various items are unrelated to each other.
- For example, a workstation may produce many parts that are unrelated but that meet some external demand requirement.
- In dependent demand, the need for any one item is a direct result of the need for some other item, usually a higher-level item of which it is part.

- For example, if an automobile company plans on producing 500 cars per day, then obviously it will need 2,000 wheels and tires (plus spares). The number of wheels and tires needed is *dependent* on the production levels and is not derived separately.
- The demand for cars, on the other hand, is *independent* it comes from many sources external to the automobile firm and is not a part of other products it is unrelated to the demand for other products.

Inventory Costs

- **A) Ordering or Setup cost over the period**

A ordering or setup is an aggregate of a fixed cost that is independent of the amount ordered, and a variable cost that depends on the amount ordered.

- It includes amongst other cost elements Clerical/labor costs of

- processing orders
- inspection
- return of poor quality products
- transport costs and
- handling costs

B) Holding costs over the period

- This is one of the vital costs that need to be optimized in any logistics system
- This inventory cost component includes the lost investment income caused by having the asset tied up in inventory.
- Storage costs, rent/depreciation, labor, overheads (e.g. heating, lighting, and security), money tied up (opportunity cost, loss of interest), and stock deterioration (lose money product due to deterioration whilst held), obsolescence costs (if left product exceeds its useful life) and insurance.

C) Shortage costs over the period

- When a customer seeks the product and finds the inventory empty, the demand can either go unfulfilled or be satisfied later when the product becomes available. The former case is called a lost sale, and the latter is called a backorder.
- **D) Purchasing Costs over the period**
- This is the unit cost of purchasing the product as part of an order.
- This include price paid on labor, material and overhead charges necessary to produce the item

- The most common inventory situation faced by manufacturers, retailers, and wholesalers is that stock levels are depleted over time and then are replenished by the arrival of a batch of new units.
- For the *basic EOQ model* to be presented first, the only costs to be considered are
 - K = setup cost for ordering one batch,
 - c = unit cost for producing or purchasing each unit,
 - h = holding cost per unit per unit of time held in inventory.
- The objective is to determine when and by how much to replenish inventory so as to minimize the sum of these costs per unit time.
- Assumptions (Basic EOQ Model)
 1. A known constant demand rate of a units per unit time.
 2. The order quantity (Q) to replenish inventory arrives all at once just when desired, namely, when the inventory level drops to 0.
 3. Planned shortages are not allowed.

Materials Requirements Planning

- Materials Requirements Planning (MRP -1) was introduced around 1960.
- this system of MRP can be used by small manufacturing organization as well to plan and control its production and inventories.

MRP System

- Out of the seven important resources (7Ms) as given below. Materials is one, which requires lot of planning and controlling in the organization. The following are the important resources, which require proper planning and management in the organization.
 - 1. Money
 - 2. Materials
 - 3. Manpower
 - 4. Machines
 - 5. Methods
 - 6. Management
 - 7. Matrix, which is system, facilities, plants and location etc

- The following things are required for this.
 - (i) The complete production structure of final product called B.O.M. explosion.
 - (ii) The lead time required to produce or purchase the different items at various levels of production system.
 - (iii) End product demand and delivery schedule as MPS.
 - (iv) The materials and parts stored as inventory ready to supply.
 - (v) The purchase orders, which can be placed to procure the items, which are not produced

- **The major objectives of MRP are to**
 - Ensure the availability of materials, components and parts for planned production and products for customers delivery
 - Maintain the lowest possible level of inventory
 - Plan manufacturing activities, delivery schedules and purchasing activities
 - To enable rapid action to be taken to overcome material or component shortages due to emergencies , late delivery ,etc

	Materials requirement planning	Economic order quantity
1	Product/ component oriented	Every item oriented
2	Dependent demand	independents demand
3	No lead time demand	Continuous lead-time demand
4	Time-phased ordering signal	Reorder point ordering signal
5	Future production base	historical demand base
6	Forecast end items only	Forecast all items
7	Quantity and time based	Quantity based only
8	Safety stock is for end item only	Safety stock is for all item

Types of Inventory controlling systems

- The various types of inventory control systems used to order and replenish the inventories are
 - (1) Two-bin system
 - (2) Maximum-minimum system
 - (3) Economic order quantity (EOQ) system
 - (4) Fixed order quantity and variable cycle or Q system
 - (5) Fixed cycle and variable quantity or P system
 - (6) Replenishment system or S,s policy
 - (7) Ordering with quantity discountsAll these systems are discussed below